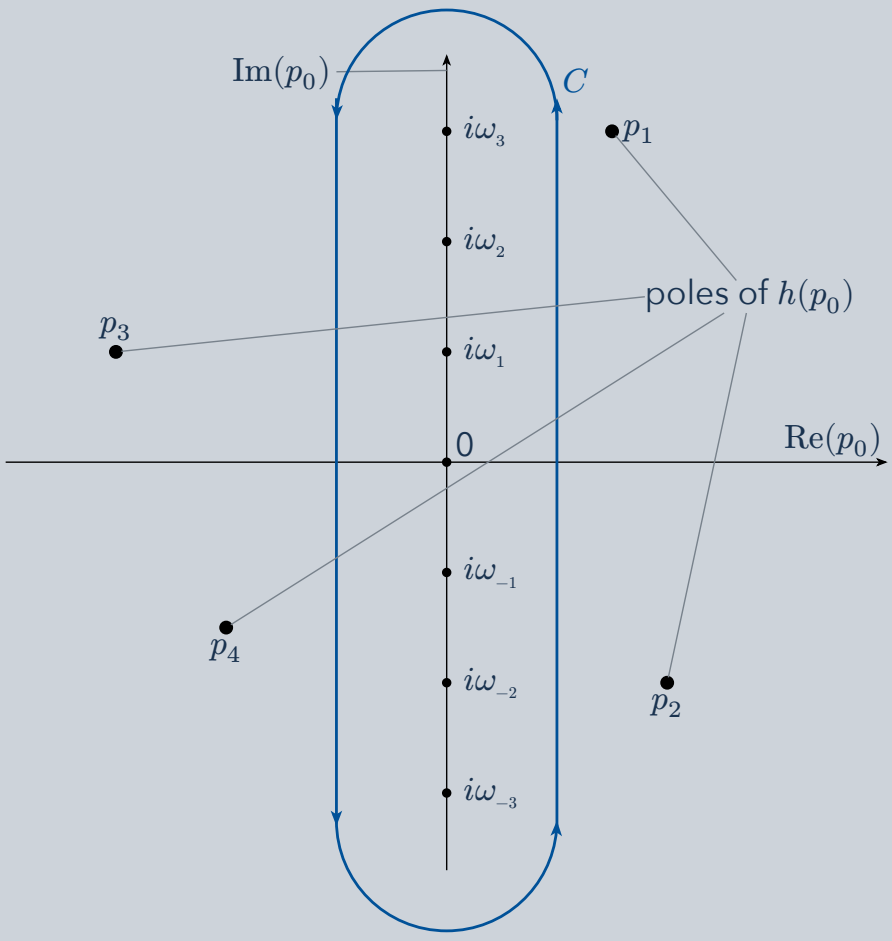
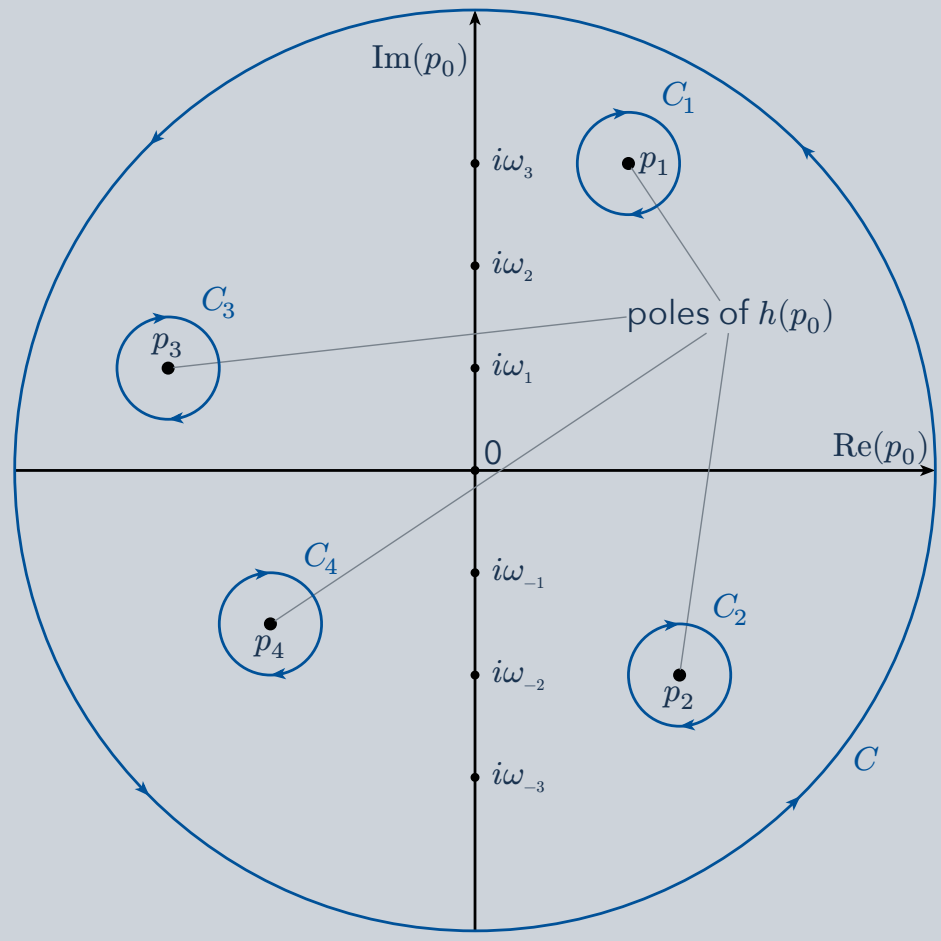
A thermal frequency sum can be rewritten as a contour integral. Follow which singularities are enclosed, and keep track of the orientation when the contour changes.

1 Enclose the thermal poles



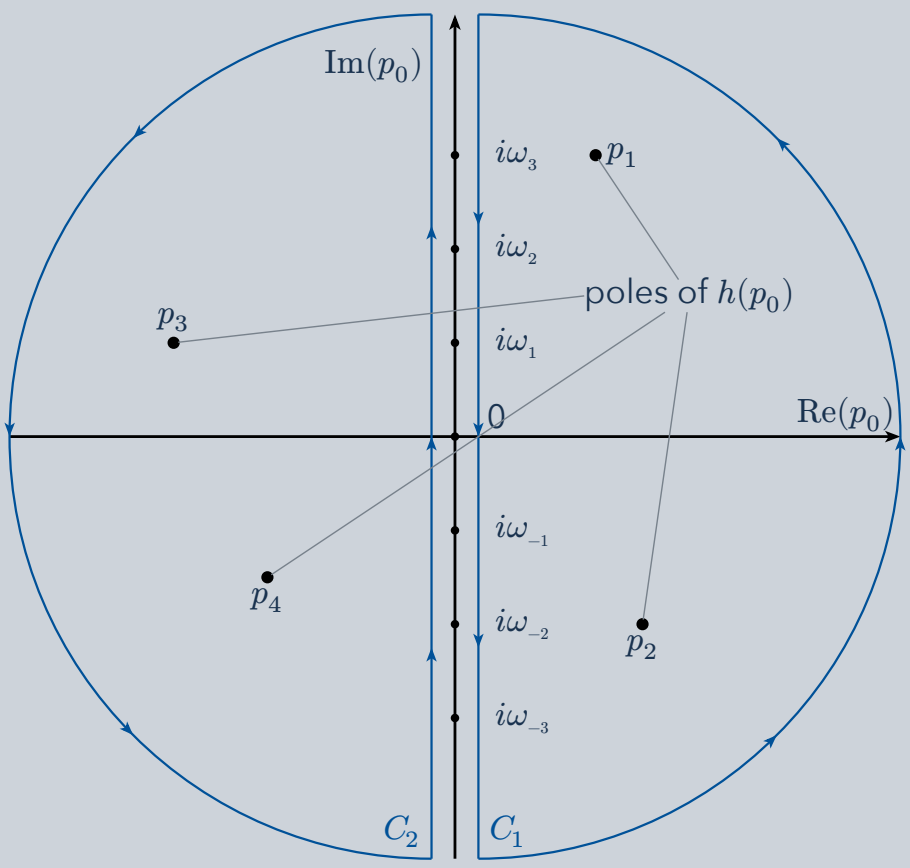
The blue contour encloses the imaginary-axis Matsubara poles of the thermal factor. Dots away from that axis are poles of the other factor, $h(p_0)$.

2 Expand, then subtract



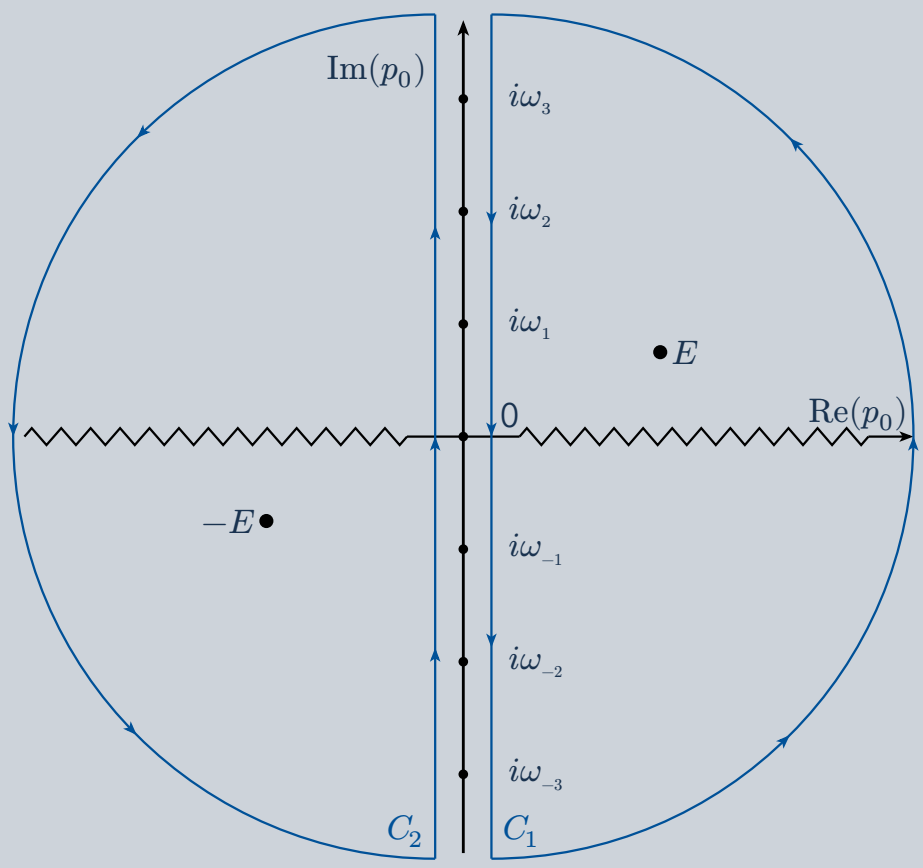
Expanding the contour also encloses poles of h. Clockwise circles subtract them back out. The outer arc vanishes only if the full integrand decays sufficiently fast.

3 Separate the half-planes



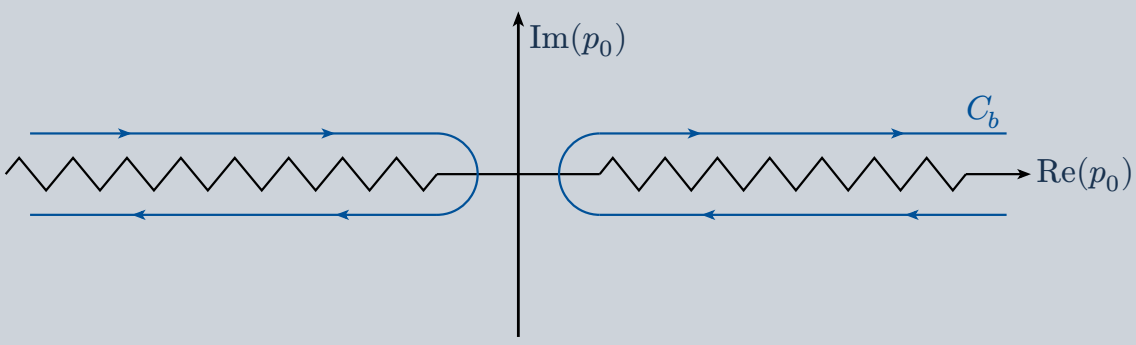
Contours C_1 and C_2 run on opposite sides of the imaginary axis. Deformation preserves the integral only while no singularity crosses the path.

4 Identify additional singularities



Here the example has poles at $+E$ and $-E$ and a real-axis cut. This is a different singularity pattern from the four-pole sketch above.

5 Follow both sides of a cut



The branch-cut contour samples the difference between the limiting values above and below the cut. A cut contributes through this discontinuity.

6 Account for every contribution

$$\sum_{n\in\mathbb{Z}} Th(i\omega_n)$$

↓

thermal-factor contour integral

↓

pole residues + cut integrals

with signs fixed by contour orientation

The pictures are a bookkeeping guide, not a universal signed formula. Specify the thermal factor, its residues, and the analytic structure of the complete integrand.

Bosons: $\omega_n = 2\pi n T$ **Fermions:** $\omega_n = (2n+1)\pi T$

Units: $k_\mathrm{B} = \hbar = 1$. Check the arc at infinity before discarding it.